Haokun Zhu

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Education

- **Carnegie Mellon University** — Pittsburgh, United States
  Ph.D. in Robotics; Advisor: Ji Zhang — *Aug. 2026 – Present*
- **Carnegie Mellon University** — Pittsburgh, United States
  Master of Science in Robotics; **GPA: 4.25/4.3** — *Aug. 2024 – Jun. 2026*
- **Shanghai Jiao Tong University** — Shanghai, China
  B.Eng in Computer Science and Technology; **GPA: 89.14/100; 3.81/4.3** — *Sep. 2020 – Jun. 2024*

Publications (* means equal contribution, ♯ means corresponding author)

- **SysNav: Multi-Level Systematic Cooperation Enables Real-World, Cross-Embodiment Object Navigation**
  Haokun Zhu, Zongtai Li, Zihan Liu, Kevin Guo, Zhengzhi Lin, Yuxin Cai, Guofei Chen, Chen Lv, Wenshan Wang, Jean Oh, Ji Zhang
  IEEE RA-L 2026. [Website] [ArXiv:2603.06914]

- **STRIVE: Structured Representation Integrating VLM Reasoning for Efficient Object Navigation**
  Haokun Zhu*, Zongtai Li*, Zhixuan Liu, Wenshan Wang, Ji Zhang, Jonathan Francis, Jean Oh
  ICRA 2026; RSS 2025 SemRob Workshop. [Website] [ArXiv:2505.06729]

- **MOSAIC: Generating Consistent, Privacy-Preserving Scenes from Multiple Depth Views in Multi-Room Environments**
  Zhixuan Liu, Haokun Zhu, Rui Chen, Jonathan Francis, Soonmin Hwang, Ji Zhang, Jean Oh
  ICCV 2025. [Website] [ArXiv:2503.13816]

- **AesStyler: Aesthetic Guided Universal Style Transfer**
  Ran Yi*,♯, Haokun Zhu*, Teng Hu, Yu-Kun Lai, Paul L. Rosin
  ACM MM 2024. [PDF]

- **SAMVG: A Multi-stage Image Vectorization Model with the Segment-Anything Model**
  Haokun Zhu*, Juang Ian Chong*, Teng Hu, Ran Yi♯, Yu-Kun Lai, Paul L. Rosin
  ICASSP 2024. [ArXiv:2311.05276]

- **M3DM-NR: RGB-3D Noisy-Resistant Industrial Anomaly Detection via Multimodal Denoising**
  Chengjie Wang, Haokun Zhu, Jinlong Peng, Yue Wang, Ran Yi, Yunsheng Wu, Lizhuang Ma, Jiangning Zhang
  TPAMI 2025. [ArXiv:2406.02263]

- **Stroke-based Neural Painting and Stylization with Dynamically Predicted Painting Region**
  Teng Hu, Ran Yi♯, Haokun Zhu, Liang Liu, Jinlong Peng, Yabiao Wang, Chengjie Wang, Lizhuang Ma
  ACM MM 2023. [Code] [ArXiv:2309.03504]

- **Phasic Content Fusing Diffusion Model with Directional Distribution Consistency for Few-Shot Model Adaption**
  Teng Hu, Jiangning Zhang, Liang Liu, Ran Yi♯, Siqi Kou, Haokun Zhu, Xu Chen, Yabiao Wang, Chengjie Wang Lizhuang Ma
  ICCV 2023. [Code] [ArXiv:2309.03729]

Experience & Internship

- **Zhang Lab in CMU** — Pittsburgh, United States
  Ph.D. Student advised by Dr. Ji Zhang — *Aug. 2026 – Present*
  - **Research Focus**: embodied AI, vision-language navigation, and real-world object navigation systems.
- **roBot Intelligence Group (BIG) in CMU** — Pittsburgh, United States
  Graduate Research Assistant co-advised by Prof. Jean Oh and Dr. Ji Zhang — *Sep. 2024 – Jun. 2026*
  - **VLM-Guided Object Navigation in Simulation**: designed structured multi-layer scene representation (viewpoint, object, and room nodes) and two-stage navigation policy combining VLM reasoning with classical exploration methods, achieving state-of-the-art performance across HM3D, RoboTHOR, and MP3D benchmarks (ICRA 2026).
  - **SysNav: Cross-Embodiment Navigation System in Real-World**: built a three-level system for building-scale object navigation, deployed on wheeled, quadruped, and humanoid robots across 190 real-world experiments with 4-5x higher efficiency than baselines (RA-L 2026).
- **Digital Media Computer Vision Laboratory(DMCV) in SJTU** — Shanghai, China
  Undergraduate Research Assistant advised by Prof Ran Yi — *Oct. 2022 – Jun. 2024*
  - **Image Generation & Stylization**: few-shot diffusion model adaption (ICCV 2023), aesthetic guided style transfer (ACM MM 2024), and SAM-based image vectorization (ICASSP 2024).

- **Youtu Lab at Tencent Technology (Shanghai) Co.Ltd** Shanghai, China
  Research Intern advised by Jinlong Peng *Dec. 2023 – Jun. 2024*
  - **Multimodal Industrial Anomaly Detection**: addressed ineffective feature integration issues in 3D point cloud and RGB images by applying multimodality to enhance industrial anomaly detection.

Research Projects

SysNav: Cross-Embodiment Object Navigation in Real-World (Robotics) *May. 2025 - Jun. 2026*

Advisor: Ji Zhang, Jean Oh

- We propose SysNav, a three-level object navigation system that decouples semantic reasoning, navigation planning, and motion control, treating rooms as minimal decision-making units and supporting both standard and conditional object navigation in building-scale real-world environments. In this project,
  - **I integrated VLM reasoning into the ROS2-based TARE Planner and deployed the system on a wheeled robot, Unitree Go2 quadruped, and Unitree G1 humanoid, validating the embodiment-agnostic design.**
  - **I evaluated the system through 190 real-world experiments, achieving 4-5x higher navigation efficiency than existing baselines.**
  - **This work is accepted by IEEE Robotics and Automation Letters (RA-L) 2026.**

STRIVE: VLM-Guided Object Navigation in Simulation (Robotics) *Sep. 2024 - May. 2025*

Advisor: Jean Oh, Ji Zhang, Jonathan Francis

- We propose STRIVE, a framework that enables robots to incrementally build a multi-layer scene representation during exploration and leverages VLM reasoning for efficient object navigation. We designed a two-stage navigation policy combining high-level VLM-guided planning with low-level exploration methods. In this project,
  - **I led the design of the structured scene representation module including viewpoint nodes, object nodes, and room nodes to ground VLM understanding in robotic navigation.**
  - **I implemented core components of the navigation framework and the two-stage policy that integrates VLM reasoning with classical exploration methods.**
  - **I conducted extensive benchmark evaluations across HM3D v1/v2, RoboTHOR, and MP3D, achieving state-of-the-art performance in both success rate (↑7.1%) and efficiency (↑12.5%).**
  - **This work is accepted by ICRA 2026 and the RSS 2025 SemRob Workshop.**

AesStyler: Aesthetic Guided Universal Style Transfer (CV) *Jun. 2023 - Dec. 2023*

Advisor: Ran Yi, Yu-Kun Lai, Paul L. Rosin

- We propose AesStyler, a novel Aesthetic Guided Universal Style Transfer method, utilizing pre-trained aesthetic assessment model, a novel Universal Aesthetic Codebook and a novel Universal and Specific Aesthetic-Guided Attention module. Extensive experiments and user-studies have demonstrated that our approach generates aesthetically more harmonious and pleasing results than the state-of-the-art methods. In this project,
  - **I proposed three novel contributions and implemented them independently, which empowers AesStyler to produce results aesthetically more harmonious and pleasing compared to current SOTA methods.**
  - **I completed the initial drafting of the research paper by myself, meticulously explaining the methodology, crafting figures to illustrate the pipelines and analyzing the experimental results in detail.**
  - **I single-handedly designed and conducted all the experiments, both qualitative and quantitative and 2 user-studies, demonstrating the superiority of AesStyler over current state-of-the-art methods.**
  - **This work is accepted by ACM MM 2024.**

SAMVG: An Image Vectorization Model with SAM (CV, CG) *Aug. 2023 - Dec. 2023*

Advisor: Ran Yi, Yu-Kun Lai, Paul L. Rosin

- We propose SAMVG, a multi-stage model to vectorize raster images into Scalable Vector Graphics. Extensive experiments demonstrate that SAMVG can produce high quality SVGs in any domain with less computation time and complexity compared to previous SOTA methods. In this project,
  - **I collaborated with the co-author to propose three innovative aspects.**
  - **I finished the final drafting of the paper, elaborating in detail on the methodology and conducted a thorough analysis of the experimental results in the paper.**
  - **This work is accepted by ICASSP 2024.**

M3DM-NR: RGB-3D Noisy Industrial Anomaly Detection (CV) *Dec. 2023 - Jun. 2024*

Advisor: Jinlong Peng

- We propose M3DM-NR, a novel noise-resistant framework to leverage the strong multi-modal(image and point cloud) discriminative capabilities of CLIP. Extensive experiments show that M3DM-NR outperforms state-of-the-art methods in 3D-RGB multi-modal noisy anomaly detection. In this project,
  - **I proposed the 3-stage noise-resistant framework for noisy industrial anomaly detection.**
  - **I single-handedly designed and conducted all the experiments, both qualitative and quantitative, demonstrating the superiority of M3DM-NR over current state-of-the-art methods.**
  - **I completed the initial drafting of the research paper by myself, meticulously explaining the methodology, crafting figures to illustrate the pipelines and analyzing the experimental results in detail.**
  - **This work is accepted by TPAMI.**

Technical Skills

- **Languages** Python, C/C++, Matlab, LaTex, Mandarin(native), English(fluent, TOEFL: 111)
- **Tools** ROS, PyTorch, TensorFlow, OpenCV, OpenGL, LaTex, Markdown, git